Software Requirements Specification

for

KADA Cooperative System

Version 1.1 approved

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Table of Contents

Revision History

Name	Date	Reason For Changes	Version
KodeLab Team	29-10-2024	Initial release of the document	1.0
KodeLab Team	09-11-2024	Add system architecture Improve use case and use case description	1.1

1. Introduction

The KADA Cooperative System is a web-based portal designed to modernize and manage membership and loan services of a cooperative. It will replace the manual, paper-based system currently in use by the cooperative, where members can apply for loans, maintain their accounts, and also be active in cooperative affairs. The system offers multiple features to members, administrators, and Board of Directors through various subsystems for membership and loan processing administration and reporting. This will be a unitary structure that fosters efficiency, openness, and confidence amongst members to participate actively in the operations and financial undertakings of the cooperative.

1.1 Purpose

This paper aims to provide the fundamentals for building the KADA Cooperative System. It outlines the objectives and limits of the system, particularly its digital features on the membership registration and loan application processes of KADA. This paper will elaborate on the functional aspects of the system, its constraints, and its way of handling the data. This paper will also serve as a user guide as well as the foundation for the system initial version in development by the team.

1.2 Document Conventions

This document uses the following conventions.

1.2.1 Numbering and structure

- Numbered list for sections and subsections (e.g., 1.0, 2.1).
- Nested numbering for detailed requirements (e.g., 1.2.1, 4.2.1).
- Normal numbering for list of features, specifications or steps (e.g., 1., 2.).

1.2.2 Text Formatting

- **BOLD:** Indicate as the title.

1.2.3 Tables and Point

- **Tables** to list requirements and Use Cases (e.g. hardware requirement).
- **Bullet Point** to list key features and specifications.

1.2.4 Document Style

- This document uses “Times New Roman”, font size 12 for content, 14 for subheadings and 18 for headings.

- The document using the 1.15 spacing.

1.3 Intended Audience and Reading Suggestions

Project Managers: The project managers have to plan out the projects, schedule, and ensure alignment with the project objectives in consonance.
Reading suggestion: can read the Overview and Scope sections for an overview of what this project is trying to achieve, then review the Functional Requirements and Intended Use sections to verify that alignment between the goals and deliverables of the project are met.

Marketing Staff: Marketing staff focuses on how to understand system mission and benefits effectively communicate to potential users.
Reading suggestion:can read the Overview to understand the mission of the system, then check the Intended Use section to identify benefits and key features that will enable you to communicate the value of the system to users effectively

End Users :

Applicants/ Members: Interact with the system to perform operations like applying for loans and knowing the status of the account.
Reading suggestion: can read the Overview and User Role Definitions sections, which outline how the system supports membership and loan processes

Administrators: Performing tasks such as account management and maintenance of user permissions and reporting functions.
Reading suggestions: can view the Administrative Functions section for information about account management and permission control, as well as reporting functions

Testers: Testers develop test cases that will help find out whether the system meets functional and non-functional requirements of the system.
Reading Suggestion: can review the Functional Requirements and User Role Definitions to understand the workflows and create appropriate test cases, then, review the Non-Functional Requirements, which capture performance, security, and usability criteria.

Documentation Writers: User manuals and guides in respect of the system shall be prepared by the documentation writers.
Reading Suggestion: can read this entire document, beginning with the Overview section, to gain a thorough understanding of the system's feature set, workflows, and technical information essential for crafting clear and complete documentation.

1.4 Product Scope

The scope of the project is to allow the cooperative members to perform services like applying for loans, managing their accounts and participating more actively in the affairs of the cooperative. As of now, the old KADA Cooperative System is still using a hard copy system for processing applications for membership and loan requests. The website for the cooperative being developed should be accessible for viewing by any person irrespective of whether one is a KADA cooperative member or not. However, the cooperative system only allows access to the members of the cooperative who can also administer the application process and see the financial reports.

The KADA Board of Directors can access the general monetary health report as prepared for KADA cooperative and they also help process members and loans through carrying out cooperational meetings. This allows for the assessment of the effectiveness of the executives responsible for the day-to-day running and helps in building the trust of the members and active participation by communicating on matters such as what the cooperative is doing, achievements as well as problems. An administrator also applies the use of the websites in managing the cooperation and giving the right information on the cooperation. The administrator can manage user information as well as check membership and loans status. Model operation of our system are stated as following:

1. Members subsystem
2. Admin subsystem
3. Board of Director subsystem
4. Membership and loan subsystem
5. Reporting subsystem

1.5 References

1. Lam, G. (n.d.). *Business Rules vs. Business Requirements : Features : Business Rules Community / Business Rules Journal*. Business Rules Community - BRC. https://www.brcommunity.com/articles.php?id=c141
2. "Software Requirements Specification." *Page has been moved*, https://www.cse.chalmers.se/~feldt/courses/reqeng/examples/srs_example_2010_group2.pdf [Accessed 2 November 2024]
3. Bandakkanavar, R., & Mohanty, S. (2023, May 8). *Software Requirements Specification document with example*. Krazytech. Retrieved October 31, 2024, from https://krazytech.com/projects/sample-software-requirements-specificationsrs-report-airline-database

2. Overall Description

The KADA Cooperative System is an online, website-based system that proposes to computerize and streamline most of the cooperative processes that require membership registration, loan applications, and other financial management, which were previously done by hand. The modules this system would employ are member modules, administrator modules, Board of Directors' modules, and reporting for easy access in core functions. Key features to include are identity authentication, online loan application, real-time profile, and finance update. It includes considerations for compliance and security in the design to ensure safe data handling and works on stable internet access with a compatible browser. Full-featured support materials round out the system so users can easily navigate the system, thereby enhancing efficiency and accessibility of all members within the cooperative.

2.1 Product Perspective

This product is a new website system for KADA Corporation which is aimed at making a complete change from the hard copy methods of processing membership and loan applications to more automated systems. It tries to help the user eliminate all the inconveniences that are associated with applying for membership and loans, managing user details and even managing reports. In addition, the system incorporates the capabilities of calculating cooperatives' loans and interests for each member, reducing the workload on the people.

In Figure 2.1, a three-layered structure under the KADA Cooperation system architecture is illustrated. The Presentation Layer is comprised of the clients who will be the applicants, KADA Cooperative Staff, BODs and KADA administrators interacting with the system through website User Interface. Next, the Business Layer is where the system elicits actions such as user authentications, member management, loan management, managing the approval statuses and generating reports. Last, the Data Layer is the component that maintains records for loans, members, and cooperative activities and supplies data for the functions of the Business Layer.

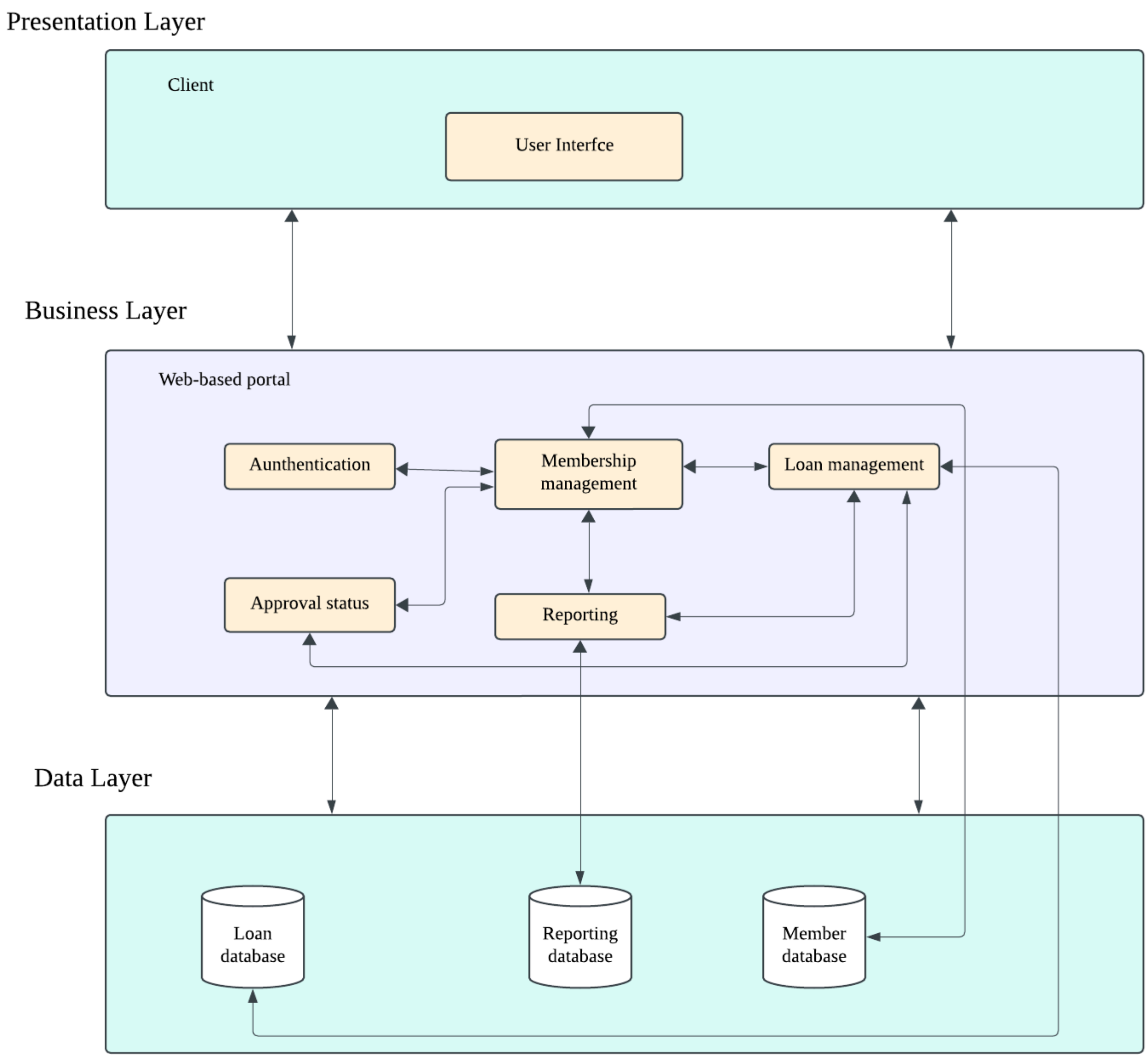

Figure 2.1 System Architecture

2.2 Product Functions

There is a few major functions that the user can perform using this system :

1. Identity Authentication

   In this system, users can verify their identity through the login feature to gain access. If a user forgets their password, they can reset it using the "Forgot Password" feature. This function also can determine their role based on the email inserted.

2. Member Registration

   KADA staff can register as cooperative members by completing an online registration form in the system. Membership is required to access member features in the system where the system will direct guests to the member registration form from “Sign Up”.

3. Loan Application

   KADA cooperative members can apply for loans online by filling out an application form in the system, eliminating the need to visit the KADA cooperative office in person by clicking on the button “Apply” on the List of Loans page.

4. Review Application

   This function allows each person in the BODs of the KADA Cooperative to view, approve, or reject both member and loan applications directly in the system, streamlining the approval process during monthly meetings.

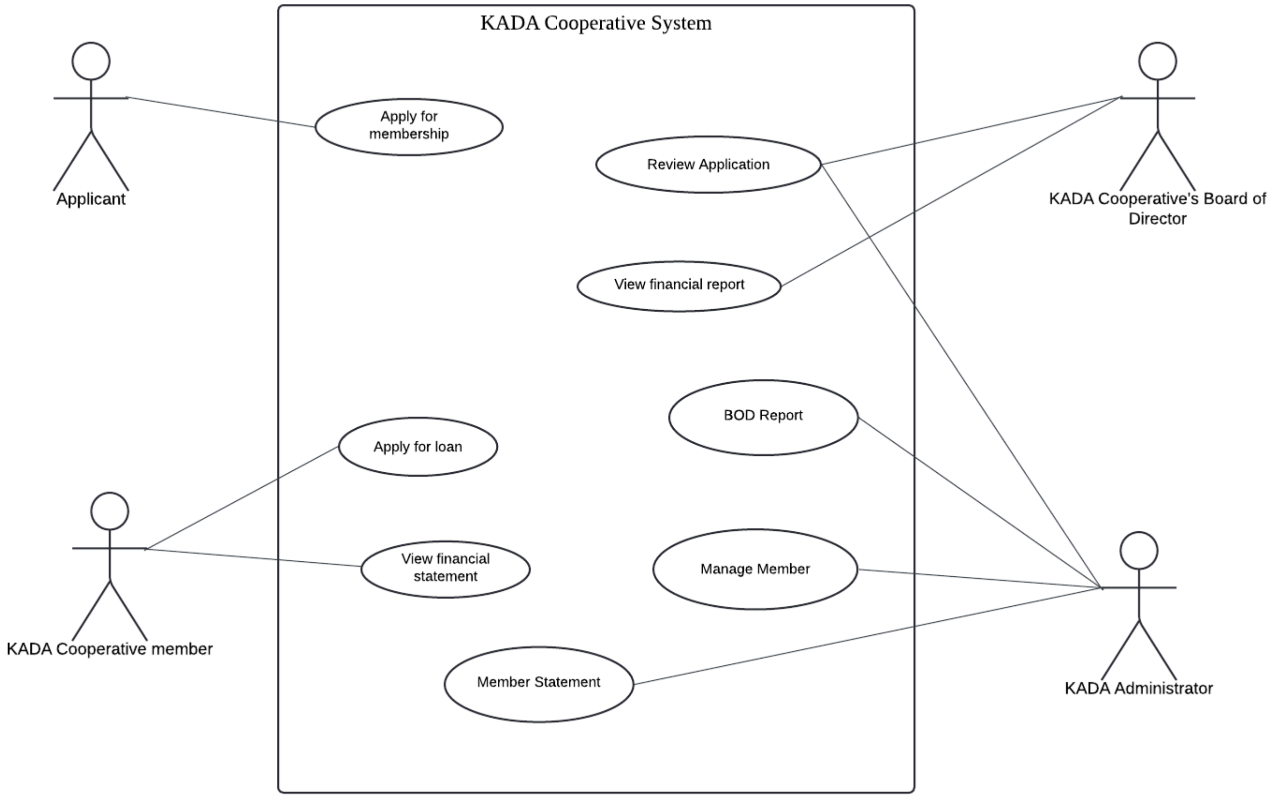

Figure 2.2: Use case diagram of the new KADA Cooperative System

2.3 User Classes and Characteristics

2.3.1 Applicant and member

- Applicant:
  These are the people applying for membership or seeking a loan. They can access the system through an online portal. Their major activities will be to fill in the membership eligibility forms and, upon acceptance, the loan application form. It lets an applicant keep themselves updated on the status of their applications in real time and search for details of particular membership and loan eligibility that best suit their requirements. This user class focuses on ease regarding the submission of the application and visibility into the application process of a loan.

- Member:
  Members include all those cooperative staff members who get accepted as members. They shall be given access to a secure portal on the web. The primary roles carried out by the members are managing their profiles, monitoring their membership status, and applying for loans if not done earlier. Members shall be informed of the available benefits and information on loans, such as type of loan, interest rate, and period of payback. The system should be able to keep them updated about their benefit status and/or any due outstanding action or renewals for or on any loans granted on date. This category emphasizes ease of

access to membership benefits and personal updates on membership and loans.

2.3.2 Administrator

The administrators play a key role in managing and maintaining the KADA system daily. They have full access to the system functionality. They can edit or update the KADA home page, review loan application statistics and check the details of membership and loan application. Administrators are responsible for approving membership and loan applications. They are allowed to view the financial records such as member statements, loan summaries, savings statements and the annual report. All of these permissions, admins require technical expertise to handle the sensitive information securely and maintain the integrity of the KADA system.

2.3.3 Board Of Directors

The Board of Directors or BOD is a significant user class of the KADA Cooperative System, with purposes of monthly meetings and decision making on a continuous basis. They may evaluate member and loan requests, retrieve the financial data, as well as check the system's metrics. The BOD member that holds high-level privileges is able to access classified content that has been passed through the verification process.

Given the wide range of technical skills possessed by members of the BOD, the system is very simple and provides helpful suggestions for them to do their duties. This model helps the cooperative run effectively, and make strategic decisions in an appropriate time frame.

2.4 Operating Environment

Table 1.0 below shows the items and description of the operating environment of the system needed.

Item	Description
Hardware Platform	- 8GB RAM and above - 256GB of storage and above - Single-core processors or multi-core processors
Operating System	- Windows OS
Database	- MySQL
Browser Compatibility	- Modern web browsers like Chrome, Firefox and Microsoft Edge.
Network Requirements	Require a stable, high-speed internet connection to ensure the data transmission between client-server highly secure.

Table 1.0 shows the operating environment of system

2.5 Design and Implementation Constraints

The KADA Cooperatives System even has its own drawbacks. The data administration, financial transactions, and reporting has to comply with the guidelines set by the Malaysian Ministry of Agriculture and Food Security. In addition, the system needs to be operational on medium range devices with low memory and processing capability, which is the case for KADA staff and board members.

In order to facilitate the smooth transition into the digital era, the new system should connect with KADA's already existing storage facilities and accounting applications, which means it has to support older data formats. The system will employ secure and centralized databases, while a web interface will be available to both mobile and desktop users accessible via standard web browsers.

Due to the significant financial value, safety is the primary factor. User authentication and data encryption will help keep unauthorized users away from the system. Furthermore, the system will follow certain development processes in order to ease the system maintenance by KADA's IT worker in the future. This includes proper documentation and programming designed to accommodate further development.

2.6 User Documentation

There are a few document components that users can refer to help them understand more about the system.

1. User Manual
    - A comprehensive manual containing system features, navigation and troubleshooting.
    - Format : PDF

2. FAQs
    - List of frequently asked questions addressing common user concerns and issues.
    - Format : HTML

3. Tooltips (Inline Help)
    - Brief descriptions and formatting tips displayed when users hover over or click a question mark or information icon near specific fields (e.g., form fields).
    - Format : Built into the user interfaces for immediate guidance.

2.7 Assumptions and Dependencies

An assumption regarding the product is that every user has to use the cooperative website with internet access. This may lead to the situation where the website has a few issues, especially when a user has a weak internet connection jeopardizing the effectiveness of the system's user experience. Another assumption is that cooperatives are financially self-sufficient and have adequate resources to meet their financial commitments at all times.

As for the dependencies, the system may depend on the third-party banks or any financial institutes for the purposes of ensuring safe transfer of funds and disbursement of loans. Besides, the system also depends on the email for the admin to send their application approval status.

3. External Interface Requirements

KADA Cooperative System is a web-based cooperative system developed to streamline the process of cooperation by providing an enabling platform where members can apply for loans, check on their application statuses, and manage their profiles online. It has key features that include modules for members, administrators, and the Board of Directors, unique in their interfaces and permissions to execute tasks in a secured and efficient manner. It shall support desktops, tablets, and payment terminals for hardware interfaces. Data storage shall be assured by MySQL. The communication interfaces through email, SMS, and secure Web access keep them updated on application improvements, among other communications, while strong security and compliance are ensured.

3.1 User Interfaces

Prior to any logins even for the members, a dashboard page as depicted in figure 2 should be the first page viewed by a guest user who has browsed the cooperative website for the first time. If the user wishes to apply for KADA Cooperative, they can go to the '?' section and complete the sign up page as shown in figure 3. After the approval of the user membership, the user can now log in, as illustrated in figure 4.

It is only the applicant for membership that can continue to the cooperative loan. Cooperatives' members that are wishing to apply for a loan can easily direct themselves to the apply for loan page illustrated in figure 5. They can access their loan application status at the status section and other details of KADA Cooperative at the dashboard.

Next, for the admin and BOD, they have their own login page as shown in Figure 6. The admin is responsible for the operation of the cooperative information including the cooperation profile, statistics, loan application, financial statement, and report generation. In the member application section, the admin will also be responsible for facilitating the review of the members' applications and their respective details by clicking on their names and modifying the member's checklist as shown in Figure 7. The administrator can also include alterations to the cooperative members' statement and make necessary changes on the annual report for the BOD who will consider it.

As indicated in Figure 8, the BOD has access to information regarding the members' application and their share models. The BOD can also see loans applied by the individual members of the cooperative and the loans that they apply for, as illustrated in Figure 9. They can also look at the finance summary of the cooperative and there are cooperative reports for every year.

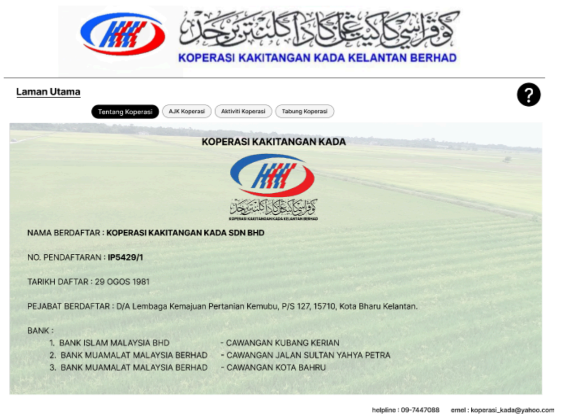

Figure 2 - Dashboard page

Figure 3 - Sign up page

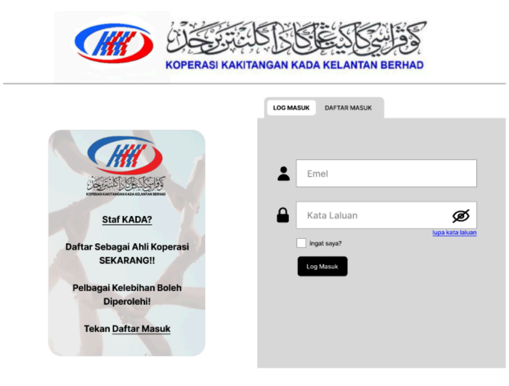

Figure 4 - Member login page

Figure 5 - Loan application

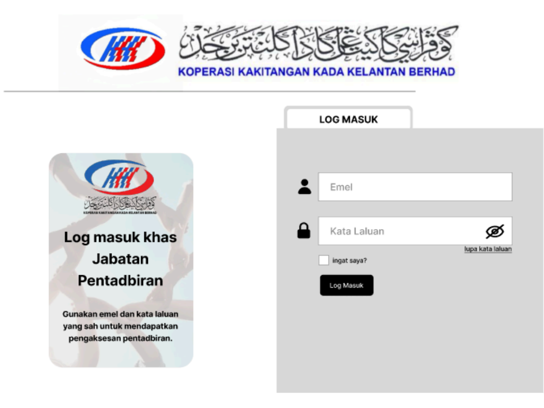

Figure 6 - Admin and BOD login page

Figure 7 - Member application (admin)

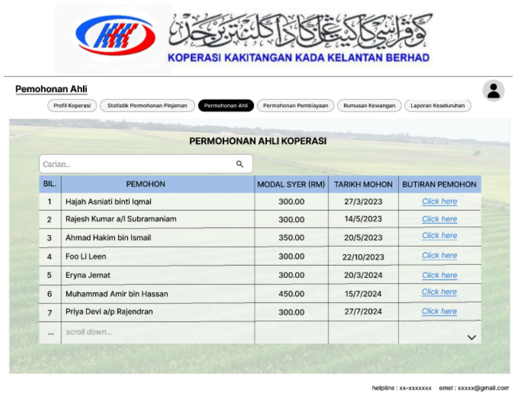

Figure 8 - Member application (BOD)

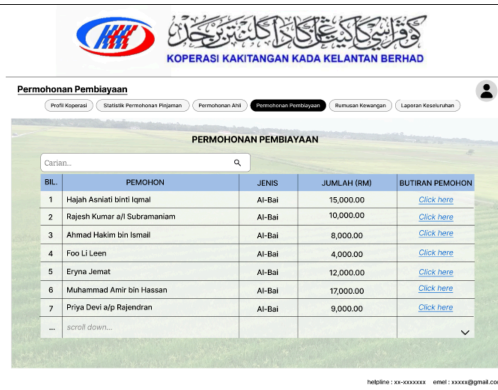

Figure 9 - Loan application (BOD)

3.2 Hardware Interfaces

These are the devices that interface with the system to ensure smooth running for the employees and board members of KADA.

a. Employee Workstations

Device Type: Desktop computers or laptops.

Data Characteristics: The employee workstations interact with the centralized database in order to perform inquiries, submissions, and updates of membership and loan application data in real time. They also facilitate the interaction with automated verification and report generation features.

Control Interactions: Workstations execute client applications that allow employees to fill in membership and loan forms, submit them for review, and view reports.

Communication Protocol: Secure data interchange using TCP/IP network protocols with the server that hosts the centralized database.

Physical Connection: The workstations will connect to the organization's secure internal network via Ethernet or Wi-Fi to ensure that they are always connected to the backend server.

b. KADA Board Member Tablets

Device Type: Tablets

Data Characteristics: These tablets will provide board members with real-time reporting, analytics dashboards, and application review tools for decision-making.

Control Interactions: The tablet interfaces allow the board members to accept and discard applications, view analytics in detail, and create reports directly from the database.

Communication protocol: HTTPS for secure web access via a secure web-browser interface.

Physical Connection: Tablets connect to the system using Wi-Fi or 4G/5G networks securely, which allows mobile access and facilitates decision-making remotely.

c. **Database Server**

Device Type: Centralized database server, it is about MySQL Server that hosts a secure repository.

Data Characteristics: The server will store and maintain vulnerable data relating to the employees' information, loan eligibility details, payment records, and financial reports on a monthly/yearly basis.

Control Interactions: The server shall handle all read/write operations, enforce encryption of data, and ensure unauthorized persons do not have access to sensitive data.

Communication Protocol: HTTPS/TLS-secure data in transit, integrating protocols for encryption of data in transit and at rest.

Physical Connection: KADA's safe server room; connection to the organization LAN via a firewall to block unauthorized access.

d. **Employee Payment Terminal Optional**

Device Type: Cash payment terminal to repay loans.

Data Characteristics: This device is used to repay loans in cash by the employee in case he does not opt for deduction from his salary. The device captures the transaction information along with reflecting the loan balance in the database.

Control Interactions: Through the application interface, the terminal will update the central database in real time regarding the payment records by processing and validating the payments.

Communication Protocol: HTTPS or Proprietary protocol is used for encrypted data exchange between Terminal and Database.

Physical Connection: The physical connection is through Ethernet or a secure Wi-Fi from the terminal to the network within KADA's premises.

This hardware interface section defines the type of hardware, data characteristics, control methods, communication protocols, and physical connection the cooperative application and loan system requires. The structure provides a good basis upon which this system should be kept secure, efficient, and user-friendly for both KADA employees and board members.

3.3 Software Interfaces

Table 1.2 shows the connections to other software components :

Item	Purpose	Description
Web Portal	Provides an interactive and responsive interface for system users.	Utilizes PHP as the main portal for accessing the KADA website.
Operating System	Provide the hosting environment for KADA Cooperative System.	The primary operating system for the server is expected to be Windows OS due to its stability and security features. For the client side, the access of the system will be compatible with Windows, macOS and Linux environments.
Database	Stores user data, member records, loan applications, transactions.	The system will use MySQL as the database management system for handling cooperative member data, financial transactions and the system logs.

Table 1.2 shows about the software interfaces

3.4 Communications Interfaces

Communication interfaces play a crucial role in the KADA cooperative system by facilitating communication between users and the system as well as between different system components.

1. Email Interface:
    - Function: Manages sending emails to the users regarding application statuses or new application updates.
    - Example: SMTP (Simple Mail Transfer Protocol) for sending emails to users regarding notifications or loan application statuses.
    - Requirements:
        - The system must support sending emails using the SMTP protocol.
        - Emails must include standard headers (To, From, Subject) and a body.
        - The system should notify users (admin) for any email delivery failures.
        - Users must have the option to opt-in for email notifications.

2. Web Browser Interface:
    - Function: Allows users to interact with the system via a web browser.
    - Example: HTTP/HTTPS protocols for web page requests and responses, ensuring secure communication.
    - Requirements:
        - The application must be compatible with major web browsers.
        - The system must use HTTP/HTTPS protocols for secure data transmission.
        - The interface needs to be adaptable to various screen sizes.

3. Database Connection Interface:
    - Function: Facilitates communication between the application and the database.
    - Example: SQL queries executed over a TCP/IP connection to fetch or store data.
    - Requirements:
        - The system must connect to the database using secure methods, such as TCP/IP.
        - SQL queries must be optimized for performance and prevent SQL injection attacks.
        - Connection pooling should be implemented to manage database connections efficiently.

4. File Transfer Protocols:
    - Function: Allows the system to upload or download files.
    - Example: FTP or SFTP (Secure File Transfer Protocol) for securely transferring financial reports or other documents into the system.
    - Requirements:
        - The system must support secure file transfers using FTP or SFTP.
        - File uploads and downloads must be tracked and recorded for security and compliance.

    - The system should implement file size limits and type restrictions to prevent abuse.
    - Users must receive confirmation of successful uploads/downloads via in-system notifications.

5. SMS Notification Interface:
    - Function: Manages sending messages to the users regarding application statuses or new application updates.
    - Example: SMS service provider that offers API to send over message.
    - Requirements:
        - Support sending brief messages (up to 160 characters).
        - Ensure data security and user consent for SMS communications.
        - Any message delivery failures must be notified by the system to the user (admin).

4. System Features

KADA Cooperative System is designed as the ultimate backbone of cooperative operations: the secure, role-based access for members, administrators, and board members. With a user-friendly login interface, membership registration, loan application, viewing financial statements, administrative management, review application this system enhances user experience, data security, and efficiency in daily operations. It manages members' profiles and applications online, administrators can track and update user information, and the board members review applications and generate reports. The KADA Cooperative System uses a protected, automated interface to support safe and efficient digitalization within the cooperatives.

4.1 System Feature 1 : Login

Use Case : Login
ID : **UC1**
Actors: 1. KADA Member 2. KADA Administrator 3. KADA BODs
Preconditions: 1. All users must have registered in the system.
Flow of events : 1. The users visit the main homepage of the system. 2. The users click on the profile icon. 3. The system displays "Login" and "Register" buttons. 4. The users click on the "Login" button. 5. The system displays the login page. 6. The users input their email and password and click "Login". 7. The system validates the data. 8. If the data is not correct 8.1 The system displays a message to try again. 9. Else 9.1 The system displays the homepage for verified KADA users.
Postcondition :
Alternative flow :

4.1.1 Description and Priority

System login tools enable the user to enter the system by their roles whether they are members or admin or BOD. It is to prevent unauthorized users from easily accessing the system and to protect confidential information such as member financial data, loan information and more. The secure login reassures users that their information and financial activities are protected which also increases the confidence in the cooperative's digital services.

- **Benefit:** 9 - Role-based access is critical.
- **Penalty:** 9 - Failing to implement the secure login system may expose the cooperative to serious risks.
- **Cost:** 5 - The costs are moderate and manageable.
- **Risk:** 7 - Require diligent management and mitigation strategies

4.1.2 Stimulus/Response Sequences

1. **Stimulus:** A user enters their email and password, then clicks the "Login" button.
   System Response: The system first validates the user.

2. **Stimulus:** The system validates the email and password successfully.
   System Response: The system initiates the user sessions and redirects the user to their respective dashboards.

3. **Stimulus:** The user enters an incorrect email or password.
   System Response: The system prompts an error message.

4. **Stimulus:** The user clicks on the "Forgot Password" link.
   System Response: The system prompts the user to enter their registered email address or phone number and will send a password reset link via email or SMS.

5. **Stimulus:** The user clicks on the "Logout" button.
   System Response: The system ends the user session and redirects the user to the login page.

4.1.3 Functional Requirements

- **REQ-1:** The system must implement user authentication by user's email id and password confirmation
- **REQ-2:** The system should allow the email ids and Passwords to be case sensitive.
- **REQ-3:** The system should not allow unauthorized access to users with invalid credentials and give an error message.
- **REQ-4:** The system should recognize different user roles when a user logs in successfully and take the user to a dashboard of his or her role.
- **REQ-5:** The system should limit users from accessing certain features based on the user role.
- **REQ-6:** The system needs to include a "Forgot Password" alternative on the login page.
- **REQ-7:** The system should provide a password reset link to the user via email or SMS.
- **REQ-8:** The system must keep the user logged in once he / she has managed to login successfully.

- **REQ-9:** The system shall enable members to log out at will by clicking the “Logout” prompt button and terminate the sessions instantly.

4.2 System Feature 2 : Member Application/Registration

Use Case : Member Application/Registration
ID : UC2
Actors: 1. Applicant 2. KADA Administrator
Preconditions: 1. The KADA Cooperative System's website can be accessed by the user. 2. The registration form had to be filled by the applicant together with all necessary notes.
Flow of events : 1. The user signs in and finds the compartment of the registration form. 2. The user fills out the form as required. 3. The user presses the submit button. 4. The system assures that their subjects are complete and true. 5. The application is assessed and the competent authority takes a decision on the acceptance or rejection of the application. 5.1 If permitted, the system creates a member id. 5.2 If not, a note containing the grounds for refusal may be recorded. 6. The system dispatches an email to the requesting user with the following content: 6.1 If Yes: contains login details. 6.2 If No: gives information on the reasons for the denial.
Alternative flow : Incomplete application: Where there is incomplete information provided, the system encourages the applicant to complete this section.
Postcondition : 1. The candidate is emailed an update regarding their application status. 2. Approved candidates get a Member ID and access to his/her account.

4.2.1 Description and Priority

Thanks to the development of the Member Application tool, KADA workers are able to send their applications for cooperative membership online instead of filling numerous forms prior to waiting for the approval. This feature is important as it quickens the registration process, reduces administrative work overload and increases member satisfaction.

- **Benefit**: 8 – Swiftly improves process and eases user experience greatly.
- **Penalty**: 7 – Without this feature, management bottlenecks and other inefficiencies exist.
- **Cost**: 5 – Development and other related costs are moderate.

- **Risk**: 3 – Low-risk primarily involves aspects of user acceptance and systems reliability.

4.2.2 Stimulus/Response Sequences

1. **User Action**: The user signs in to the system and goes to the page where registration is done.

   System Response: The registration page where membership details are to be filed is presented.

2. **User Action**: Staff input the required details and proceeds to the submission of the application stage.

   System Response: Checks the information and stores it within the database; otherwise, the user is verbally prompted to enter corrections if an error is detected.

3. **User Action**: Admin reviews the request and takes appropriate action by changing the application status to 'Approved' or to 'Rejected'.

   System Response: An employee is updated by a status message issued as an email

4. **User Action**: Log in using the credentials provided for members who have been approved.

   System Response: Member-level functionality is made available.

4.2.3 Functional Requirements

- **REQ-1**: A registration form must be provided by the system with details for personal data.
- **REQ-2**: The system should validate the information entered by the user to check that it is complete and in the correct form.
- **REQ-3**: The system should keep all membership applications that have been verified in a protected database.
- **REQ-4**: The system shall provide functionality for administrative users to view applications and change their statuses.
- **REQ-5**: The system should be capable of sending out emails to employees when there are changes on the status of applications.
- **REQ-6**: Every candidate who has been approved will be given a distinct Member ID, which will be held in a safe manner.
- **REQ-7**: The system must issue log in details to new members to enable them access members only parts of the system.

4.3 System Feature 3 : Loan Application

Use Case : Loan Application
ID : UC3
Actors: KADA Cooperative Member
Preconditions: The user must be a registered KADA Cooperative Member.
Flow of events : 1. Members select the "Loan Application" option on the system interface. 2. System displays the loan application form with required fields. 3. Members fill in all the necessary details and submit the application form. 4. System validates the application details for completeness and accuracy. 5. System shows the successful send of the form pages. 6. System saves the application data and provides a confirmation with a unique reference number.
Alternative flow : - Incomplete information : If the user leaves the required fields blank, the system displays an error message asking the user to complete the information before submit.
Postcondition :
Alternative flow : - Invalid information : If data entered is invalid or does not follow the formatting, the system prompts the user to correct the data.
Postcondition : The loan application form is submitted and the user awaits the successful status.

4.3.1 Description and Priority :

The feature enables users to submit a loan application through the system by providing necessary personal information and financial details. This feature is crucial to the loan process as it was the first step for the loan request. This step also ensures either the applicant is eligible or not to apply for the loan. Priority : High

- **Benefit** : 8 - Allow members to apply for loans conveniently and efficiently through an automated system.
- **Penalty** : 9 - Missing this feature leading to dissatisfaction among members and potential financial losses.
- **Cost** : 6 - Cost is manageable given its importance and impact.
- **Risk** : 5 - Poses moderate risk including data entry errors, incomplete information and potential system failures during application process.

4.3.2 Stimulus/Response Sequences :

1. **User Actions :** Cooperative member selects "Loan Application" on the tab above the KADA System dashboard.
   System Response : The system displays the loan application form.
2. **User Actions :** Cooperative members fill in the required information.
   System Response : The system validates the entries for completeness and accuracy.
3. **User Actions :** The user submits the application form.
   System Response : System displays the successful application and saves the application data.

4.3.3 Functional Requirements :

- **REQ-1 :** The system should display the loan application form when the user clicks the "Loan Application" button.
- **REQ-2 :** The system should require users to complete all the required fields inside the form.
- **REQ-3 :** The system must validate each field to check that the data is in the correct format or not.
- **REQ-4 :** The system should provide error messages if the user incomplete or incorrectly formatted the input fields and prompt the user to correct the errors before submitting.
- **REQ-5 :** After submitting the loan application form, the system should save the application data in the database and generate a reference number.
- **REQ-6 :** The system must notify the user of successful completion and display the references number for future use.

4.4 System Feature 4 : Member Statement

Use Case : Member Statement
ID: UC4
Actors: KADA Member (cooperative members)
Preconditions: The user has been able to log into the system successfully and has provided valid credentials.
Flow of events: 1. The user accesses the system using his or her credentials and selects the “Member Statement” section. 2. The Member specifies the type of statement they wish to see (for instance, statement of savings balance or statement of loan balance). 3. The relevant statement is retrieved and displayed on the system with the latest available data. 4. If required, the Member is provided with an option to save the statement in any format as per his/her wish (for instance, in PDF format).
Alternative flow: Data Not Found: In the event that the system encounters a challenge in retrieving data, it will inform the User and suggest to him/her to try again later.
Postconditions: When requested, the financial statement of the member is submitted or made available for download.

4.4.1 Description and Priority

Member Management will provide the facility of the members managing their profiles, loan applications, and view financial statements relating to their membership. A core functionality that allows members to have a self-serving, easy-to-use interface to manage their membership tasks, hence improving their overall experience and efficiency of the system.

- **Benefit**: 9 - Improves member engagement and system effectiveness.
- **Penalty**: 8 - If it's not developed, for example, the processes stay manual and really hard for the members.
- **Cost**: 6 - Reasonable cost to develop; hence, the grade is average.
- **Risk**: 5 - Low risk if deployed in a secure way.

4.4.2 Stimulus/Response Sequences

- **User Action**: Members of the website update any profile information.

  System Response: System updates this new information into the database and reflects this change immediately into the member's account.

- **User Action**: The member applies for a loan.

  System Response: The system provides the interface to apply for the loan; it should present the fields to populate the loan details along with related eligibility criteria.

- **User Action**: Members request their current financial statement.

  System Response: The system fetches and shows the current financial data of the member, including loan balances, savings, and recent transactions concerning their membership.

4.4.3 Functional Requirements

- **REQ-1**: Profile Management
  - Members should be allowed to view and make changes to their personal details, contact information.
  - The system shall confirm any changes in the profile for accuracy, for example, right format for a phone number, and the change shall be saved at once.
- **REQ-2**: Loan Application Interface
  - Members shall be provided with an interface to apply for a loan whereby they shall fill out loan details.
  - The system should highlight the relevant eligibility information. The system shall validate loan data, including loan amount and loan purpose, before form submission.
- **REQ-3**: View Financial Statement
  - The members should have a financial statement displaying their current standing as regards loans taken, savings, and recent transactions.
  - This statement should change dynamically upon new changes in members' financial activities with the cooperative.
- **REQ-4**: Notification of Loan and Financial Updates
  - Any changes in loan applications or financial status shall be informed to its members via the system.
  - The notifications shall be indicated on the member's dashboard and via email notification, if so desired, on the condition that the member has subscribed to such notifications.

4.5 System Feature 5 : BOD Reporting

Use Case : BOD Reporting
ID: UC5
Actors : KADA Board of directors (BOD)
Preconditions : The board member successfully accessed the system with the appropriate login information.
Flow of events : 1. The BOD Member signs into the system and navigates to the section with reports. 2. The BOD Member picks the type of report which they wish to look at (e.g. report on members, report on loan approvals). 3. The system generates the requested report with the latest information available and displays it. 4. The report can be filtered and sorted by the BOD Member as per his/her needs. 5. The BOD Member decides whether to save the report or to export it (for instance in PDF, excel etc).
Alternative flow : Report Filter Unavailable: In the event that certain data or filter is unavailable, the system alerts and advises the BOD Member of alternative filters or views.
Postconditions : The BOD Member presents the requested report or downloads it in the mode they have selected.

4.5.1 Description and Priority

The BOD Reporting tool is designed to deliver real-time information on memberships, loans, savings as well as other financial activities. It merges data to help with making informed decisions, applications inspection, and analysis of the corporation's performance. This system is essential for clarity, speed and effectiveness in making decisions, as well as observation.

- **Benefit**: 9 - Ensures strategic resolving of issues and control execution.
- **Penalty**: 8 - Without this capability, decision-making would be based on old or absent information.
- **Cost**: 6 - With the requirement of collecting and securing data in real-time, the expense is on the higher side.
- **Risk**: 4 - Data security and integrity are at the Medium risk level.

4.5.2 Stimulus/Response Sequences

- **User Action**: The BOD Member accesses the system and chooses the reporting segment.

  System Response: A dashboard loaded with options of different types of reports is displayed.

- **User Action**: A member of the BOD chooses one of the reports, for example, Loan Approval Summary.

  System Response: The system generates the relevant report to the user's request with current data and with the possibility of filtering and sorting the results.

- **User action**: A BOD member exports or downloads the report for further consideration.

  System Response: The report can be downloaded in the following formats: pdf, excel, etc.

4.5.3 Functional Requirements

- **REQ-1**: The solution must be equipped with a visual interface where one can also see data and the different selections for report generation at the same time.
- **REQ-2**: The system should be able to compile information on such matters as membership approval, loan approval, and financial overviews among other issues.
- **REQ-3**: The system should facilitate the issuance of reports with specific parameters like dates or approval status filtered or sorted.
- **REQ-4**: The system must permit the members of the Board of Directors to save different reports in the pdf and excel formats.
- **REQ-5**: The system has to ensure that all submitted reports are accessible only by the BOD members who have any credentials designated for them.
- **REQ-6**: The system shall not take too long before the reports are updated with the latest information available.

4.6 System Feature 6 : Admin Management

Use Case : Manage Member
ID : **UC6**
Actors: KADA Admin
Preconditions: The user must be an authorized KADA Administrator with appropriate system access.
Flow of events : 1. Admin logs into KADA System with their specified email and password. 2. System grants access to the admin page, providing options for editing and monitoring the system, managing the data and user management. 3. Admin views the details of the applicants of the membership and loan applications. 4. Admin selects a specific task such as editing the home page, managing the application statuses, editing the statement or updating the reports. 5. System performs the requested task and confirms the successful completion.
Alternative flow : Unauthorized Access : If an unauthorized user attempts to access the administration dashboard, the system denies the access and logs the attempt.
Postcondition : Admin tasks are performed successfully to ensure the system integrity and availability.

4.6.1 Description and Priority

The System Administrator function allows the administrator to manage and sustain various elements in KADA Cooperative System including user profiles, the status of the members and their loan requests. This functionality also helps in ensuring proper maintenance of the data within the cooperative, that is accurate, current and managed in a safe environment. It is an essential component in the day to day functioning and reliability of the system as a whole.

- **Benefit**: 9 - Facilitates effective user and information handling leading to smooth functioning of the system and security.
- **Penalty**: 7- Service management is very critical to the system that low accuracy, certainty and user friendliness may be experienced.
- **Cost**: 5 - The outlays are fairly bordering on the average though there is the need to put in place measures of protecting the data employed.
- **Risk**: 4 - The risks are also fair, dealing mostly with safety and protection of the information.

4.6.2 Stimulus/Response Sequences

1. **User Action**: The user who has administrative privileges accesses the system and goes to the Admin Dashboard.
   System Response: The screen presents options for managing users and memberships as well as processing loan applications.
2. **User Action**: For the purpose of editing or removing users the administrator clicks the “Manage Users” option.
   System Response: The system lists its users and allows viewing their accounts, editing these accounts, or deleting the accounts when needed.
3. **User Action**: The administrator activates “Review Membership Applications” and operates on all the applications in the queue.
   System Response: The system allows the administrator to approve and disallow membership applications with the provision such that he or she can write comments concerning the rejection.
4. **User Action**: The Administrator selects "Loan Applications" in order to assess, accept, or change the loan status.
   System Response: The suitable screen is presented displaying details on loan application, thus providing the chance for the administrator to change the status of the loan and notify applicant.

4.6.3 Functional Requirements

- **REQ-1**: The system should incorporate an admin dashboard that is capable of supporting user management, membership management and loan management functionalities.
- **REQ-2**: It should also enable the administrator to perform the locked down functionalities of adding, editing or removing user accounts.
- **REQ-3**: The administrator should also have the ability to either approve or disapprove new membership applications with the option to add comments where necessary.
- **REQ-4**: The system should also enable users to view and approve loan requests, and also change the status of the loan.
- **REQ-5**: The system should restrict access to administrative controls and information to only those administrators that have been granted such access.
- **REQ-6**: An alert must be issued to the applicants whenever there is a change regarding their membership status or loan application status.

4.7 System Feature 7 : Review Application

Use Case : Review Application
ID: UC7
Actors: KADA Board of Director
Preconditions: 1. BODs are logged on to the system.
Flow of events: 1. BODs go to the list of application pages. 2. The system displays a list of applications. 3. include (Find Application) 4. If BODs click on applicant's name 4.1 The system displays application details in a popup size. 5. BODs approve or reject application by clicking "Approve" or "Reject" button 6. The system display message indicates the decision is successful. 7. BODs click "OK" to close the popup.
Postconditions:
Alternative flow: 1. At any point, BODs can move to different pages.
Postconditions:
Alternative flow: 2.
Postconditions:

4.7.1 Description and Priority

The review application feature is a function where the BODs can review, access, approve, or reject both member and loan applications. This function enables the BODs to review applications efficiently and faster since the BODs can have access to application details by clicking on the applicant's name only.

- **Benefit**: **9** - Improve BODs process or meeting to review the applications.
- **Penalty**: **6** - The review application still can be done manually without the system.
- **Cost**: **8 -** It might be costly because this process includes email/SMS connections to send application status to applicants and might be using SMS services provider API.
- **Risk**: **7 -** The risk is quite high since all the data in the application must be true

and is not tampered with to make sure the decision-making process is genuine.

4.7.2 Stimulus/Response Sequences

1. **User Action**: The BODs click on "Membership Registration" or "Loan Application" on the tabs section.

   System Response: Displays the list of applications.

2. **User Action**: The BODs click on the applicant's name.

   System Response: Displays the application details in a medium popup size.

3. **User action**: The BODs click on the "Approve" or the "Reject" button.

   System Response: Display message indicates decision is successful or not.

   System Response: Send an email/SMS to the applicant containing the application status.

4.7.3 Functional Requirements

- **REQ-1**: The system should allow the BODs to view all the application details.
- **REQ-2**: The system should allow the BODs to approve or reject an application using on-screen buttons.
- **REQ-3**: The system should allow the BODs to close the application detail by clicking "X" or outside of the popup area.
- **REQ-4**: The system should allow the BODs to move to other pages.
- **REQ-5**: The system should notify the BODs whether the decision is successfully recorded or not.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The KADA Cooperative System is designed to manage membership and loan applications efficiently, with strong performance, security, and accessibility. It allows for only role-based access in which the administrator and board members can add, approve, and view applications, while members have access to personal information and the ability to apply for a loan. It assures high availability, data accuracy, secure real-time transaction, backup, and recovery while allowing the integration of payroll systems without much hassle for loan repayments.

5.1.1 System Authentication

ID: QR1
TITLE: Quick authentication and data retrieval.
DESC: The system has to authenticate the user and load the corresponding role-based dashboards to the user within a 2 seconds time span. In addition, retrieval of member information and loan application should take not more than 3 seconds to give a friendly usage to the users.
RAT: Very important in keeping the user engaged and satisfied.

5.1.2 System Scalability

ID: QR2
TITLE: System scalability
DESC: The system must allow for 1,000 users to be active at the same time and process data related to 100,000 members with no serious drop in system performance.
RAT: Important in preparing for future expansion and meeting users' phenomenal needs.

5.1.3 System Availability

ID: QR3
TITLE: System availability
DESC:The system is expected to work 100% of the time with only 8 hours of the downtime allowed for a year to enable users to have uninterrupted access.
RAT: Essential in ensuring uninterrupted service and maintaining user confidence.

5.1.4 Transaction

ID: QR4
TITLE: Transaction processing capacity
DESC: The system will be required to process transactions at a rate greater than 50 TPS during busy periods without any performance degradation.

RAT: Needed for the efficient functioning of the system under peak load conditions.

5.1.5 Data Accuracy

ID: QR5
TITLE: Data accuracy and data consistency
DESC: The system will ensure that all updates are corrected in real time and will have the data locked for a maximum 1 second in case of any changes.
RAT: Needed to maintain confidence in the management of data and interactions with the users.

5.1.6 Load Handling

ID: QR6
TITLE: Load management and endurance to stress
DESC: The system shall suspend a user load increase for an additional 40% above to a typical peak load while maintaining the performance of the system.
RAT: Useful for availability during inordinate use spikes.

5.1.7 Security Efficiency

ID: QR7
TITLE: Security efficiency
DESC: System performance should not be impaired by any security features, particularly encryption, which should not add more than 10% on response times.
RAT: Important in maintaining the security level expected as well as the user experience satisfaction.

5.1.8 Optimization

ID: QR8
TITLE: Optimized resource usage
DESC: Normal operating conditions should see the system average CPU and memory usage not exceeding 75% in order to facilitate efficient operations.
RAT: Important in avoiding any performance constraints from occurring.

5.2 Safety Requirements

Since the BOD Report Generation function provides details that contain sensitive financial and membership data, this will be strictly controlled to prevent unauthorized access to information, data breaches and misuse of personal information.

5.2.1 Data Encryption

All data passing over the system, including reports and login details, shall be protected against unauthorized access by encryption using AES-256.

5.2.2 Access Control

The reporting functionality will only be available for the authorized BOD members to access with the right login details. Multi-Factor Authentication (MFA) should be implemented so that even allowed users can access the confidential reports only after verification of their identity.

5.2.3 The Role of Audit Trails and Logging

The system should maintain detailed records of every access as well as actions taken using the reporting capabilities. Doing so would allow for tracking and examination of any questionable or unauthorized activity and assist in providing evidence in case of breach of data protection.

5.2.4 Standard security audits

By the industry standard, security audits of the system shall be carried out on a quarterly basis to identify vulnerabilities in the reporting systems and address them. In order to preserve the integrity of the system, any findings during the audit will be addressed immediately.

5.2.5 Respect for Data Protection Laws.

The BOD Reporting feature must meet any data protection laws, such as the Personal Data Protection Act (PDPA) of Malaysia provisions for the protection of personal data.

5.2.6 Data Backup and Recovery

In order to prevent loss of data in case of a crashed system or cyber sabotage, all the reports and critical information must be stored daily in a different safe place away from the main system. There should be provision in the system for restoration of data within the shortest time possible in case it is lost due to disastrous occurrences.

5.2.7 Safety Certifications

The system shall comply with the requirements of ISO/IEC 27001, the international standard for information management systems, which encompasses the implementation of measures to safeguard confidential information.

5.3 Security Requirements

1. Authorization
- The system should require only users with authority to have access and be able to manage the system data.
- The RBAC should be implemented to allow only administrators and BOD to view all the system data but the members can only access their own data.

2. Data Integrity
- The system should be able to make sure all the data in the system remains accurate and is not being tampered with.

3. Access Control
- The system should have restrictions on who can perform certain actions or view certain data within the system.
- Only administrators can do CRUD activities on the system data and the BOD can have access to approving the application through the system.

5.4 Software Quality Attributes

This section provides the quality attributes that are important in realizing the functionality, reliability, and usability of the KADA Cooperative Application and Loan Application system. This shall be done in line with KADA's interest in an effective, accessible, and correct application management-solution for both employees and board members.

5.4.1 Adaptability :

The system shall be such that it could be easily adapted to possible future changes in the policies of KADA on its cooperatives on the limit of loans, eligibility, and membership benefits.

Measure: The policy parameters updates, such as the interest rate on loans or the period for repayments, should be changeable by administrators through a settings interface at any time without any system downtime and within 10 minutes.

5.4.2 Availability :

The need: The variety of KADA staff and board members who might access the system at any one given time means that it has to be highly available especially during working hours that is from 8.00 AM - 6.00 PM.

Measure: System uptime during business hours shall be no less than 99.8%, given there is not more than 5 minutes of actual downtime per month outside of maintenance windows. In the event of unplanned failure, automatic failover mechanisms restore operations in under 10 minutes.

5.4.3 Correctness :

Requirement: Calculations of eligibility checks, amount of loan granted, interest accrual, and payment schedule shall be done correctly to support fairness and full transparency for the processing of employee applications.

Measure: Financial and eligibility calculations should have zero error tolerance. This shall be ensured by performing end-to-end testing on 100 loan applications. The success rate in test cases should be at least 99.9% related to financial accuracy.

5.4.4 Flexibility :

The system shall provide web access to all KADA employees and mobile access to the board members, considering that board members may need to approve or review applications remotely.

Measurement: Optimized for at least three classes of devices: desktop, tablet, and smartphone, with a consistent user interface across all supported form factors. Compatibility testing should be able to ensure 100% expected performance for the essential features on each of the platforms.

5.4.5 Interoperability

Requirement: The system will be able to integrate with payroll systems at KADA for the deduction of loan repayment for those employees who choose payroll as their means of paying loans.

Measurement: Integration with payroll system shall be performed correctly and has to finish each transaction in less than 5 seconds, as confirmed through stress testing done under peak load conditions.

5.5 Business Rules

The section of the KADA Cooperative System highlights the key roles and responsibilities in which the members of KADA Cooperative can view their own information and submit the loan applications. Admin has full access and manages users and reports. Admin also can approve loans and the financial officer handles payments and reports. Operational principles including inside the KADA Cooperative System are loan approvals managed by the admin, real-time transaction recording and admin approval membership. This can secure access controls by using unique logins, verification for sensitive actions and session timeouts.

6. Other Requirements

The system complies with the Electronic Transactions Act 1999 in providing legally valid electronic signatures for digital contracts, while compliance with the Personal Data Protection Act of Malaysia secures personal data through appropriate handling, user consent, and data retention policies.

6.1 Legal Requirements

A. Electronic Signature Compliance
   - The system must support electronic signatures in compliance with the Electronic Transactions Act 1999, ensuring electronic contracts are legally binding.
   - If the system requires users to sign documents digitally (e.g., in loan applications), it must comply with electronic signature laws to make these agreements legally enforceable.

B. Data Protection and Privacy Laws
   - The system must comply with the Personal Data Protection Act (PDPA) of Malaysia to ensure that all personal data collected, processed, and stored within the system is handled responsibly and securely.
   - This law might cover user consent for data collection, data retention policies, and users' rights to access or delete their data.

Appendix A: Glossary

This document use the following terms and abbreviations:

Term	Descriptions
OS	Operating System
KADA	Kemubu Agricultural Development Authority
MFA	Multi-Factor Authentication
PDPA	Malaysia’s Personal Data Protection Act
SMS	Short Message Service
ID	Identification
QR	Quick Response

RAT	Required Action Type
CPU	Central Processing Unit
DESC	Description
TPS	Transaction Processing System
REQ	Requirement
HTTP/HTTPS	HyperText Transfer Protocol / HyperText Transfer Protocol Secure
FTP/SFTP	File Transfer Protocol / Secure File Transfer Protocol
API	Application Programming Interfaces
SMTP	Simple Mail Transfer Protocol
PHP	PHP: Hypertext Preprocessor

Appendix B: Analysis Models

Appendix C: To Be Determined List